An elementary improvement of the average sensitivity bound for polynomial threshold functions

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Abstract

We refine the weighted Fourier-grading argument of OpenAI for average sensitivity on the uniform Boolean cube. For every polynomial threshold function of degree at most d in n variables, we obtain $\mathrm{AS}(f) \le 3.83\, d\sqrt{n}$, improving the coefficient 8 in that argument. More precisely, if $b_n = \mathbb{E}|B_n - n/2|$ for $B_n \sim \mathrm{Bin}(n, 1/2)$, then $\mathrm{AS}(f) \le \min\{n, 8db_n + 2d^2, 10db_n\}$. The improvement follows from separating the two directions of displacement in a coupling with equal marginals and a one-sided displacement constraint. The proof is uniform in dimension and degree, permits zeros of the defining polynomial with the positive sign convention, and yields an asymptotic coefficient $8/\sqrt{2\pi}$ when $d = o(\sqrt{n})$.

Keywords: polynomial threshold functions; average sensitivity; Boolean functions; coupling; Gotsman–Linial conjecture; Fourier grading.

1 Statement and relation to previous work

Write $\Omega_n = \{-1, 1\}^n$ and let X be uniform on Ω_n. For $x \in \Omega_n$, let $x^{\oplus i}$ denote x with coordinate i reversed. For $f : \Omega_n \to \{-1, 1\}$, its average sensitivity is

$$\mathrm{AS}(f) = \sum_{i=1}^{n} \mathbb{P}\{f(X) \neq f(X^{\oplus i})\}. \tag{1}$$

We use $\mathrm{sgn}(t) = 1$ for $t \ge 0$ and $\mathrm{sgn}(t) = -1$ for $t < 0$.

OpenAI [1] establishes $\mathrm{AS}(f) \le 8d\sqrt{n}$ for a real degree-at-most-d polynomial threshold function. Its weighted-grading argument reduces the estimate to a coupling of binomial random variables. The present note retains that reduction and sharpens the probability step. We give the finite-dimensional reduction in Section 3, following [1, Lemma 3 and Section 5], to make the dependence of the numerical coefficient explicit.

Earlier general estimates include those of Harsha et al. [2], Diakonikolas et al. [3], and Kane [4]. Kane's bound has the form $\sqrt{n}(\log n)^{O(d \log d)} 2^{O(d^2 \log d)}$. Kabanets, Kane, and Lu [5] give another proof of the optimal dimension exponent. The stronger conjecture identifying exact symmetric maximizers is false [6, 7]; the result below concerns a uniform upper bound, not exact maximizers. Recent results of Chang, Slote, Volberg, and Zhang [8] and of Tseng and Volberg [9] concern fractional sensitivity moments and retain logarithmic factors at moment order one. Grebennikov [10] proves related influence bounds for low-degree rational functions and polynomial nonvanishing indicators. Those classes differ from arbitrary polynomial sign functions.

Theorem 1. *Let $n \geq 1$ and $0 \leq d \leq n$ be integers, let p be a real polynomial of degree at most d, and put $f = \operatorname{sgn} p$ on Ω_n. Then*

$$\mathrm{AS}(f) \leq \frac{383}{100}\, d\sqrt{n}. \tag{2}$$

With

$$b_n = \frac{n}{2^n}\binom{n-1}{\lfloor (n-1)/2 \rfloor} = \mathbb{E}|B_n - n/2|, \qquad B_n \sim \mathrm{Bin}(n, 1/2), \tag{3}$$

the following stronger parameter-dependent estimate holds:

$$\mathrm{AS}(f) \leq \min\{n,\ 8db_n + 2d^2,\ 10db_n\}. \tag{4}$$

2 A refined one-sided coupling estimate

Lemma 2. *Let U, V have the same distribution, finite variance, and common mean m. Suppose $V - U \leq a$ almost surely, where $a \geq 0$, and put $b = \mathbb{E}|U - m|$. Then*

$$\mathbb{E}(U - V)^2 \leq \min\{4ab + a^2,\ 5ab\}. \tag{5}$$

Proof. Define $D = V - U$ and

$$g(t) = \int_m^t |z - m|\, dz = \tfrac{1}{2}(t - m)|t - m|.$$

For $u < v$, minimizing this integral over intervals of length $v - u$, and integrating the chord bound for $|z - m|$, respectively, give

$$\frac{(v-u)^2}{4} \leq g(v) - g(u) \leq \frac{v-u}{2}(|u - m| + |v - m|). \tag{6}$$

The lower bound can also be checked directly by separating intervals that cross m from those contained in one half-line. Since g has quadratic growth, the finite second moment makes $g(U), g(V)$ integrable. Equal marginals imply

$$\mathbb{E}[(g(U) - g(V))\mathbf{1}_{D<0}] = \mathbb{E}[(g(V) - g(U))\mathbf{1}_{D>0}].$$

On $D < 0$, the lower bound in (6) controls D^2 by four times the corresponding potential increment. On $D > 0$, its upper bound and $D \leq a$ control that increment. Consequently,

$$\begin{aligned}
\mathbb{E}[D^2\mathbf{1}_{D<0}] &\leq 4\mathbb{E}[(g(U) - g(V))\mathbf{1}_{D<0}] \\
&= 4\mathbb{E}[(g(V) - g(U))\mathbf{1}_{D>0}] \\
&\leq 2a\mathbb{E}[(|U - m| + |V - m|)\mathbf{1}_{D>0}] \leq 4ab.
\end{aligned}$$

The bounded direction satisfies $\mathbb{E}[D^2\mathbf{1}_{D>0}] \leq a^2$, giving the first estimate. Alternatively, $\mathbb{E}D = 0$ and the triangle inequality give

$$\mathbb{E}D_+ = \tfrac{1}{2}\mathbb{E}|U - V| \leq b, \qquad \mathbb{E}[D^2\mathbf{1}_{D>0}] \leq a\mathbb{E}D_+ \leq ab.$$

This proves the second estimate. Independence and exchangeability are not required. □

3 The weighted-grading reduction

This section records the reduction from [1] with the normalization in (1). All matrix norms and dimensions below are finite-dimensional.

Let $N = 2^n$ and use the counting inner product on $\mathbb{C}^{\Omega_n}$. For $S \subseteq [n]$, put $\chi_S(x) = \prod_{i \in S} x_i$ and let V_k be the span of the characters with $|S| \le k$. Fix a positive weight w, let D_w be multiplication by $\sqrt{w}$, and set

$$K_k = D_w V_k, \qquad E_k = K_k \cap K_{k-1}^{\perp}, \qquad K_{-1} = \{0\}.$$

Let Π_k be the projection onto E_k. Since D_w is invertible, $\dim E_k = \binom{n}{k}$, and the centered grading operator

$$M = \sum_{k=0}^{n} (k - n/2)\Pi_k$$

is self-adjoint and has $\|M\|_{\mathrm{HS}}^2 = nN/4$. If Z_i is multiplication by x_i, then

$$\Pi_s Z_i \Pi_r = 0 \qquad \text{for } |s - r| > 1. \tag{7}$$

Indeed, Z_i commutes with D_w and raises polynomial degree by at most one; self-adjointness gives the opposite triangular vanishing.

Proposition 3 (Edge recovery, following OpenAI)**.** *For $h : \Omega_n \to \{-1, 1\}$, let H be multiplication by h and let E_h be the set of ordered neighboring pairs (x, y) with $h(x) = h(y)$. Then*

$$|E_h| \le 2\|MH + HM\|_{\mathrm{HS}}^2. \tag{8}$$

Proof. Put $C_i = [M, Z_i]$. Define the unitaries $J = \sum_k (-1)^k \Pi_k$ and $W = \sum_k \mathrm{i}^k \Pi_k$. By (7), $O_i = (Z_i - JZ_iJ)/2$ consists of the adjacent-grade blocks of Z_i. Thus $WO_iW^* = \mathrm{i}C_i$ and $\|C_i\|_{\mathrm{op}} \le 1$. In the point basis $(C_i)_{xy} = (y_i - x_i)M_{xy}$, so $|M_{xy}| \le 1/2$ for $x \ne y$. Also, (7) gives $\|C_i\|_{\mathrm{HS}}^2 \le N$. Writing ρ for Hamming distance,

$$\sum_{x,y} \rho(x, y)|M_{xy}|^2 = \tfrac{1}{4} \sum_i \|C_i\|_{\mathrm{HS}}^2 \le \tfrac{nN}{4} = \sum_{x,y} |M_{xy}|^2.$$

Let $A_0 = \sum_x |M_{xx}|^2$ and $F = \sum_{\rho(x,y) \ge 2} |M_{xy}|^2$. The preceding inequality gives $F \le A_0$. Since there are nN ordered neighboring pairs,

$$\sum_{\rho(x,y)=1} (1/4 - |M_{xy}|^2) = A_0 + F \le 2A_0.$$

Every summand is nonnegative. With $A_h = \sum_{(x,y) \in E_h} |M_{xy}|^2$, we therefore have $|E_h|/4 \le A_h + 2A_0 \le 2(A_h + A_0)$. The entries of $T = MH + HM$ are $T_{xy} = (h(x) + h(y))M_{xy}$. Its squared norm is at least $4(A_h + A_0)$, proving (8). □

Proposition 4 (Binomial coupling, following OpenAI)**.** *For $f = \operatorname{sgn} p$ of degree at most d, there is a coupling U, V of $\mathrm{Bin}(n, 1/2)$ such that $V - U \le d$ and*

$$\mathrm{AS}(f) \le 2\mathbb{E}(U - V)^2. \tag{9}$$

Proof. Multilinearization preserves the values of p on the cube without raising its degree. By adding a sufficiently small positive constant, one can also ensure that p never vanishes while preserving f under the stated sign convention. Put $\chi = \chi_{[n]}$, $w = |p|$, and $h = \chi f$. Equal-sign neighbors for h are precisely the sensitive neighbors for f, so $\mathrm{AS}(f) = |E_h|/N$.

The Fourier support of $q = \chi p$ lies in degrees at least $n - d$. For $a \in V_r$ and $b \in V_s$,

$$\langle D_w b, H D_w a \rangle = \sum_x \overline{b(x)} a(x) q(x) = 0 \qquad \text{if } r + s < n - d.$$

Here $\bar{b}a$ has Fourier degree at most $r+s$. Hence $\Pi_s H \Pi_r = 0$ in this range. Since H is a self-adjoint unitary, the nonnegative weights

$$\mathbb{P}(R = r, S = s) = N^{-1} \|\Pi_s H \Pi_r\|_{\mathrm{HS}}^2$$

have both marginals $2^{-n}\binom{n}{r}$. The blocks of $T = MH + HM$ satisfy $\Pi_s T \Pi_r = (s + r - n)\Pi_s H \Pi_r$, giving

$$N^{-1} \|T\|_{\mathrm{HS}}^2 = \mathbb{E}(R + S - n)^2.$$

Now put $U = S$ and $V = n - R$. Binomial symmetry preserves the marginal law, and $R + S \geq n - d$ gives $V - U \leq d$. Applying Proposition 3 proves (9). □

4 Extracting the uniform constant

Applying Lemma 2 to Proposition 4 with $a = d$ gives $\mathrm{AS}(f) \leq \min\{8db_n + 2d^2, 10db_n\}$. Together with $\mathrm{AS}(f) \leq n$, this proves (4).

For completeness, set $c_k = 4^{-k}\binom{2k}{k}$. Summing the upper half of the binomial law and using $j\binom{n}{j} = n\binom{n-1}{j-1}$ yields

$$b_{2k} = kc_k, \qquad b_{2k+1} = (2k+1)c_k/2,$$

and hence (3). The inequality

$$c_k \leq (3k+1)^{-1/2} \qquad (k \geq 0) \tag{10}$$

follows by induction from $c_0 = 1$, the ratio $c_{k+1}/c_k = (2k+1)/(2k+2)$, and the identity

$$(2k+2)^2(3k+1) - (2k+1)^2(3k+4) = k \geq 0.$$

For even $n = 2k$, this gives $8b_n/\sqrt{n} \leq 4\sqrt{2k/(3k+1)} \leq 4\sqrt{2/3}$. For odd $n = 2k+1 \geq 15$, the decreasing ratio $(2k+1)/(3k+1)$ gives $8b_n/\sqrt{n} \leq 4\sqrt{15/22}$. Thus, for every $n \geq 15$,

$$\frac{8b_n}{\sqrt{n}} \leq K := 4\sqrt{15/22}. \tag{11}$$

Let $C = 383/100$. If $d = 0$, then f is constant. For $d \geq 1$, first suppose $n \leq C^2 d^2$. In this case $\mathrm{AS}(f) \leq n \leq Cd\sqrt{n}$. Otherwise $n > C^2 d^2 \geq 14.6689$, so $n \geq 15$ and $d/\sqrt{n} < 1/C$. Equations (4) and (11) give

$$\frac{\mathrm{AS}(f)}{d\sqrt{n}} < K + \frac{2}{C} < C.$$

The final comparison holds exactly: $C - 2/C > 0$, $K^2 = 120/11$, and

$$(C^2 - 2)^2 - \frac{120}{11}C^2 = \frac{524329931}{1100000000} > 0.$$

This proves Theorem 1. Finally, Stirling's formula gives $b_n = \sqrt{n/(2\pi)}(1 + O(1/n))$; inserting this in (4) yields the asymptotic coefficient stated in the abstract.

Remark 5. The value 3.83 is a convenient sufficient uniform coefficient. It is not asserted to be optimal. The parameter-dependent bound (4) can be substantially smaller in particular regimes. The estimates and the final rational comparison are analytic; numerical coupling and matrix experiments are not used to prove the theorem.

5 Verification and discussion

The universal bound follows from the analytic lemmas and case split above. For reproducibility, Supplementary Material S1 (`supplementary-verification.zip`) contains the exact-arithmetic script `verify_127.py`, its recorded output, and supplemental coupling and matrix experiments. The exact checks verify the rational certificate, binomial identities, the central-binomial induction inequality, and the resulting estimate through dimension 2000. This covers 14,584 nontrivial dimension–degree pairs and 1,986,416 pairs in the trivial branch, together with 3,240 rational interval checks.

The supplemental numerical checks comprise eight coupling linear programs and 63 weighted-grading examples. Their floating-point output is exploratory and is not used as a premise in the proof. The coefficient 3.83 is sufficient uniformly; the more precise estimate (4) can be smaller when d is small relative to $\sqrt{n}$. This note does not claim an optimal coefficient, exhaustive confirmation of historical priority, or a Lean formalization of the refinement.

Use of artificial intelligence. OpenAI Codex assisted with deriving the probability refinement, preparing the exposition, conducting literature searches, and generating and checking exact-arithmetic and numerical verification scripts. The separate proof audit was also AI-assisted and is not an independent human referee report. The weighted-grading construction and edge-recovery argument are due to OpenAI [1]. The full argument is provided so that the refinement can be examined independently.

Declarations

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Data and code availability. No empirical datasets were collected. The verification scripts and recorded exact and numerical outputs are included in Supplementary Material S1. The exact script uses only the Python standard library; optional numerical checks use NumPy and SciPy. Reproduction instructions and software versions are provided in the supplementary archive.

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